Binary Classification & Matching

STAT606 ASSIGNMENT

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Agenda

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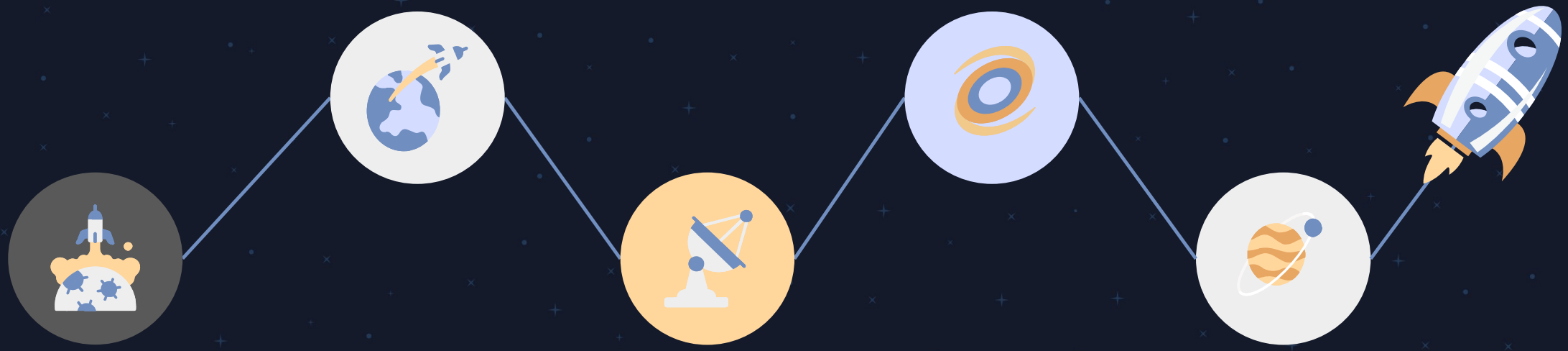


Context

In recent years, the study of Near-Earth Objects (NEOs) has become increasingly important as scientists and space agencies seek to identify astronomical objects which may pose a threat to Earth.

This project uses data from NASA's "Objects Near Earth" dataset, which catalogues 24 000 asteroids/objects, observed over the last century, and includes several astronomical and physical characteristics of asteroids such as orbital parameters, observation history, and estimated size.

The main objective of this analysis is to build a binary classification model to predict whether an asteroid is potentially hazardous (TRUE) or not (FALSE).

Predicting this accurately is crucial because the consequences of overlooking a hazardous asteroid (false negative) could be severe.

This project aims to contribute to a more reliable early-warning system for planetary defense initiatives.

Data Exploration & Cleaning: Feature Engineering

No.	Features
1	id
2	neo_reference_id
3	name
4	name_limited
5	designation
6	absolute_magnitude_h
7	is_sentry_object
8	kilometers_estimated_min
9	kilometers_estimated_max
10	orbit_id
11	orbit_class_type
12	perihelion_distance
13	aphelion_distance
14	first_observation_date
15	last_observation_date
16	orbit_class_description

Unique identifiers, names and long text descriptions were dropped

No.	Features
1	absolute_magnitude_h
2	is_sentry_object
3	kilometers_estimated_min
4	kilometers_estimated_max
5	orbit_class_type
6	perihelion_distance
7	aphelion_distance
8	first_observation_date
9	last_observation_date

avg_diameter = (min+max)/2

orbital_range = aphelion_dist – perihelion_dist

observation_duration_days = last_obs_date – first_obs_date

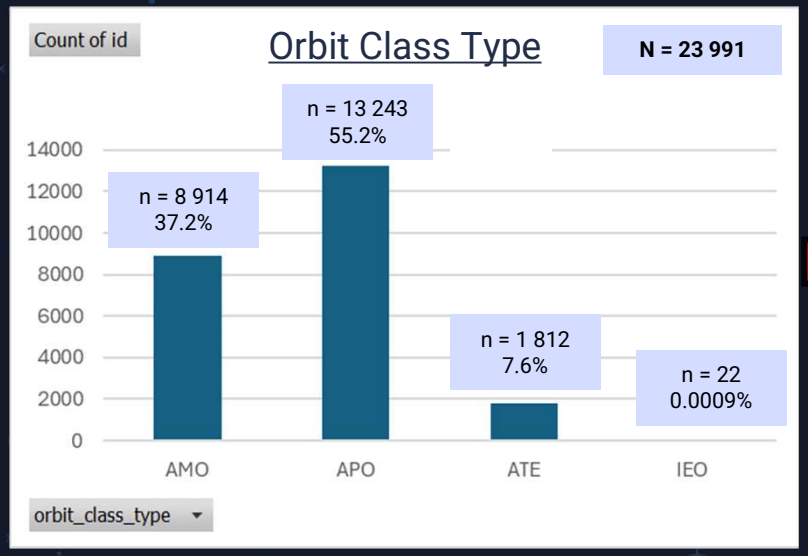
Additional Data Cleaning Done:

- absolute_magnitude_h had **8 missing values**, these were **removed**.
- observervation_duration_days had one invalid value, this was because the first observation_date was the same as the last_observation_date. The team inferred that this was a rare asteroid as it has only been seen **once** in the last 100 years, The team chose to **remove** this observation as well.

Data Exploration & Cleaning: Feature Engineering

Further feature engineering was performed on the categorical variable, orbit_class_type:

Code	Name	Description
AMO	Amor	Orbits are **outside Earth's orbit** but come close. They **approach but do not cross** Earth's orbit
APO	Apollo	Earth-crossing orbits. Most hazardous asteroids fall into this class
ATE	Aten	These objects spend most of their time **inside Earth's orbit** but still **cross it.**
IEO	Atira	Orbits are **entirely inside Earth's orbit.** These are rare and hard to detect. Sometimes called "Inner Earth Objects" or Atira asteroids



Data Cleaning:

Since Atira asteroids are rare and only represent 0.0009% of our dataset, we combined these observations with Aten asteroids, which also fall within earth's orbit. This helps reduce dimensionality. Additionally, we renamed our new categories so they are easier to interpret.

Code	New Value	n (%)
AMO	Earth_crossing_outer	8 914 (37.2%)
APO	Earth_crossing_inner	13 245 (55.2%)
ATE + IEO	Near_earth	1 834 (7.6%)

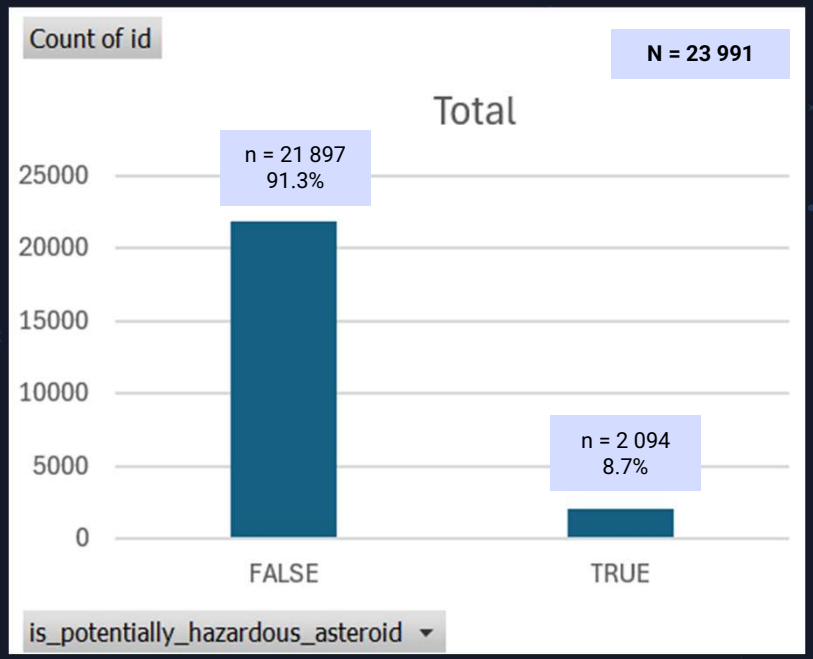
Data Exploration & Cleaning: Feature Engineering

Variable Type	No.	Features	Description	Data Type
Feature	1	absolute_magnitude_h	The **absolute magnitude** (brightness) of the object. It's a measure of how bright the object would appear at a standard distance. Lower values = brighter object; higher values = dimmer object.	Numeric
	2	is_sentry_object	A flag indicating whether the object is being monitored in NASA's **Sentry Risk Table** — a list of objects with a chance of Earth impact.	Boolean (TRUE/FALSE)
	3	avg_diameter	Size of the object	Numeric
	4	orbit_class_type	Describes the object's **orbital classification**	Categorical
	5	orbit_range	The range between the **farthest** and **closest distance** (in AU — Astronomical Units) between the object and the Sun during its orbit.	Numeric
	6	observation_duration_days	No. of days that the asteroid has been observed by NASA	Numeric
Response	7	is_potentially_hazardous	Indicates if the asteroid was detected as potentially hazardous or not	Boolean (TRUE/FALSE)

Data Exploration & Cleaning: Target Variable Analysis

1 Target Variable Analysis

Analysing the distribution of the target variable, we get:



Potentially hazardous asteroids (TRUE) are the minority class at 8.7%, therefore the team will investigate balancing techniques for the test data

2 Selecting Balancing Technique

Analysing the performance of the test & training set, with log regression being our base model of comparison, we get:

	technique	dataset	accuracy	sensitivity	specificity	precision	f1	auc
1	train_under_data	train	0.8973134	0.9391045	0.8555224	0.8666667	0.9014327	0.9163972
2	train_under_data	test	0.8513964	0.9451074	0.8424298	0.3646409	0.5262458	0.9040457
3	train_over_data	train	0.8873730	0.9351524	0.8395936	0.8535848	0.8925089	0.9047263
4	train_over_data	test	0.8541059	0.9427208	0.8456269	0.3688142	0.5302013	0.9033209
5	train_smote_data	train	0.8884290	0.9365224	0.8403357	0.8543457	0.8935487	0.9057940
6	train_smote_data	test	0.8545227	0.9403341	0.8463119	0.3692596	0.5302826	0.9031295
7	train_both_data	train	0.8883389	0.9374538	0.8405469	0.8512087	0.8922520	0.9045093
8	train_both_data	test	0.8563985	0.9427208	0.8481388	0.3726415	0.5341447	0.9029818
9	train_NASA	train	0.9105403	0.1194030	0.9861856	0.4524887	0.1889466	0.9045223
10	train_NASA	test	0.9085035	0.1121718	0.9846997	0.4122807	0.1763602	0.9020084

- The team assessed undersampling, oversampling, a combination of over and undersampling and SMOTE techniques to balance the data and compared this to how the standard train/test set split performed when no balancing was applied.
- The key metric the team used to assess performance is sensitivity and AUC (more on this later).
- Although undersampling slightly outperformed other methods by a narrow margin, we selected SMOTE due to its ability to retain more data and avoid the loss of informative examples

Note

- Assuming an asteroid **is not** hazardous when it **is** (FN) would be significantly worse than assuming an asteroid **is hazardous** when it **is not** (FP). Therefore, the team has selected **sensitivity** as a key metric to assess model performance.

- The other metric the team will be using is AUC i.e., the model's ability to distinguished between two classes.

- Naïve Bayes will not be used as there is most likely class conditional dependence between features such as orbit_class_type (orbit classification) , is_sentry (whether the object is being monitored in a list of objects with a chance of Earth impact) etc.

- Therefore, the team will be using the following four ML methods:
  - Logistic Regression Model
  - Decision Tree
  - Super Vector Machine
  - Neural Network

- A seed value of 123 was used.

- 10-fold cross validation was used.

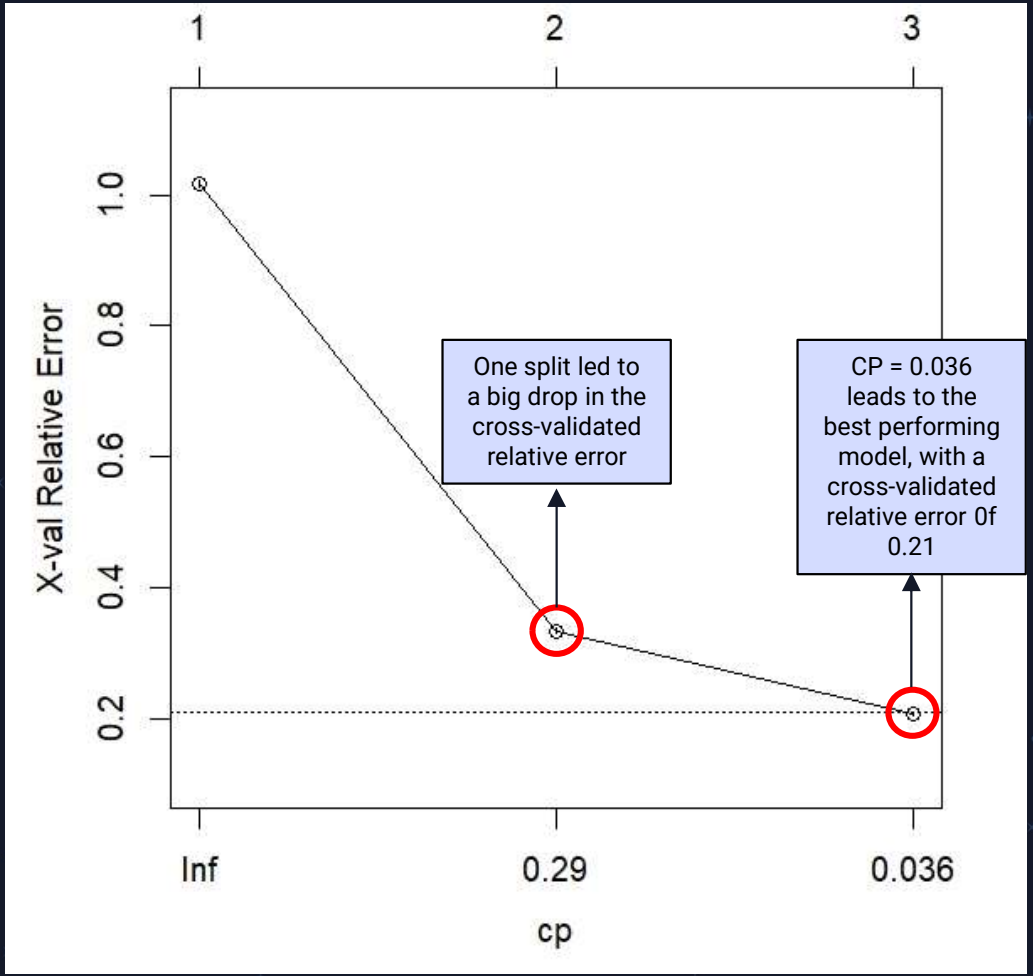
- The team used a train/test split of 80/20 as specified in the assignment instructions.

- A threshold value of 0.5 was used.

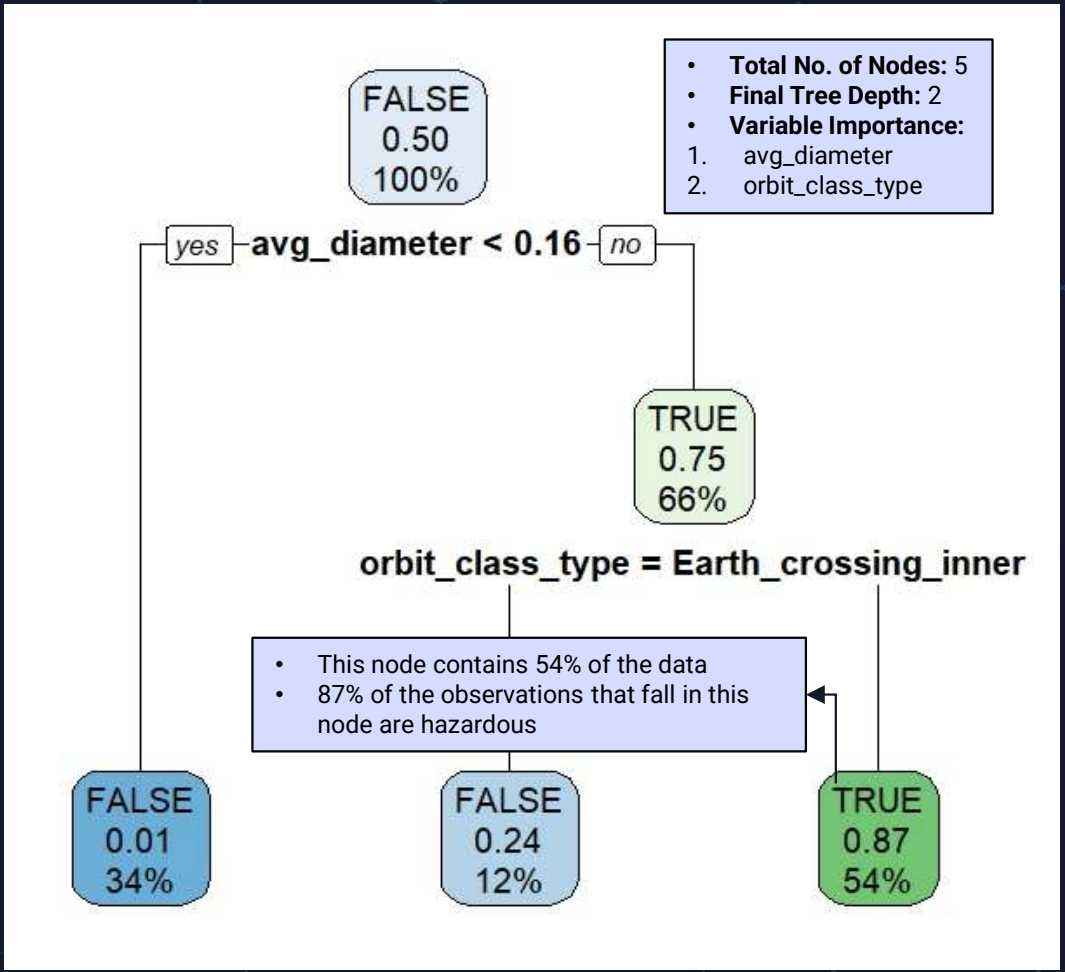
Decision Tree

Small asteroids (avg_diameter < 0.16) are almost always **non-hazardous**, while **larger asteroids** could be hazardous depending on their orbital class type

Complexity Parameter



Fitted Tree



Logistic Regression

With this ML technique, we can observe the **absolute_magnitude_h is also a feature of importance**. Large asteroids tend to be less hazardous on average. The decision tree did a better job at picking up that large asteroids are hazardous depending on orbital class type

Coefficient	Estimated Value	Interpretation	p-value	Statistically significant?
$\hat{\beta}_{\text{intercept}}$	0.23	N/A	0	Yes
$\hat{\beta}_{\text{orbit_range}}$	0.002	For every 1 unit increase in orbit_range (range in distance between object and sun), there is a 0.002 times increase in the log odds of an asteroid being hazardous.	0.487	No
$\hat{\beta}_{\text{is_sentry_object_TRUE}}$	-0.37	Objects being monitored on NASA's Sentry list have lower odds of being hazardous than ones that are not.	0.049	Yes
$\hat{\beta}_{\text{observation_duration_days}}$	0.00006	For every 1 unit increase in observation duration_days (no. of days the asteroid has been observed), there is a 0.00006 times increase in the log odds of an asteroid being hazardous.	0	Yes
$\hat{\beta}_{\text{absolute_magnitude_h}}$	-1.12	For every 1 unit increase in absolute_magnitude h (i.e., the more dim an object is), there is a 1.12 times decrease in the log odds of an asteroid being hazardous.	0	Yes
$\hat{\beta}_{\text{avg_diameter}}$	-2.74	For every 1 unit increase in avg_diameter (size), there is a 2.74 times decrease in the log odds of an asteroid being hazardous.	0	Yes
$\hat{\beta}_{\text{orbit_class_type_Near_Earth}}$	2.91	Objects **near earth** have significantly **higher odds** of being hazardous than objects **within earth's orbit**.	0	Yes
$\hat{\beta}_{\text{orbit_class_type_Earth_crossing_outer}}$	3.14	Objects **outside earth's orbit** have significantly **higher odds** of being hazardous than objects **within earth's orbit**.	0	Yes

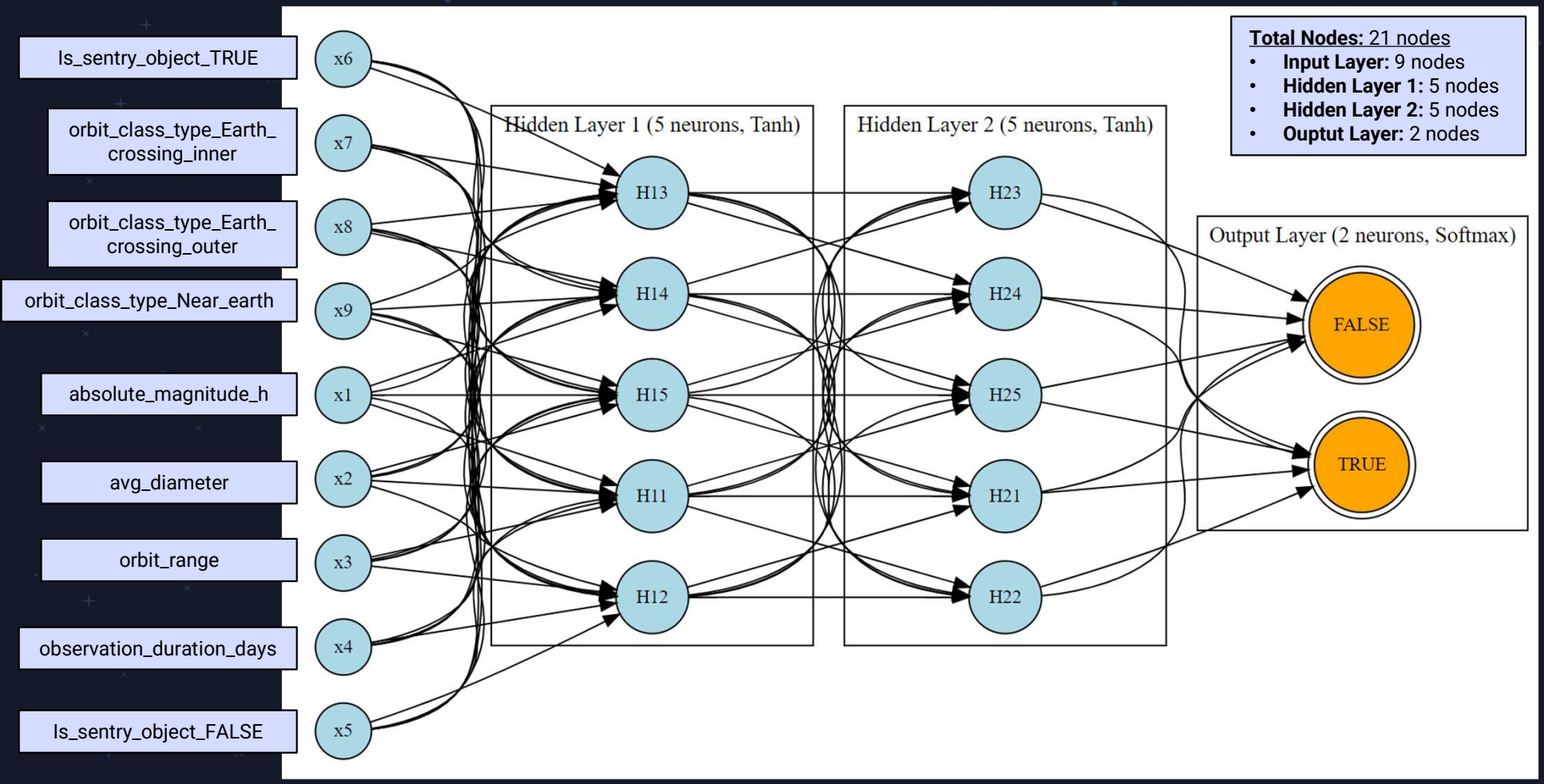
Support Vector Machines

Therefore, **SVM_poly** will be our model of comparison moving forward

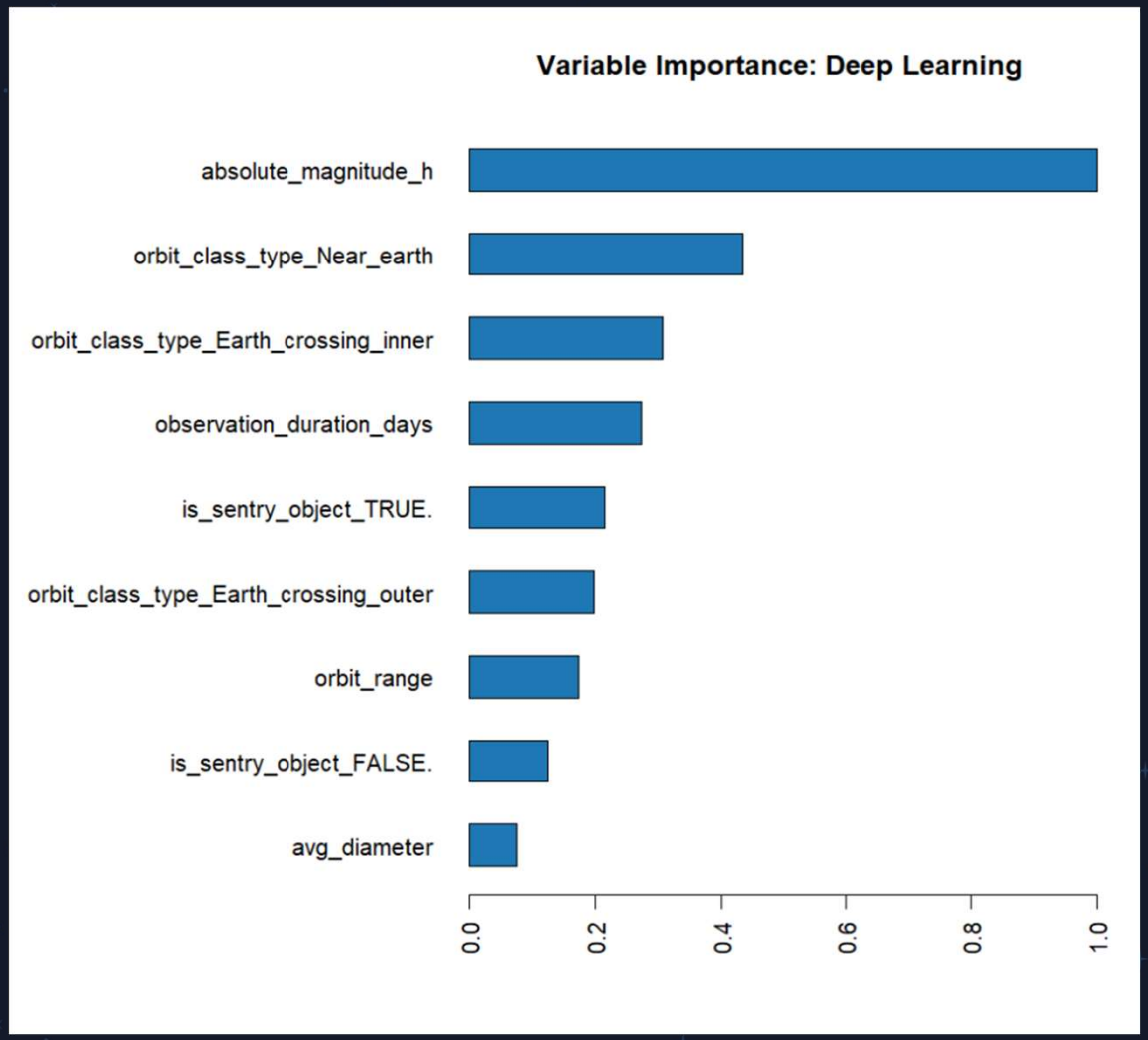
	2 SVM_linear	1 SVM_poly	3 SVM_radial
Kernel Function	$K(x_i, x_j) = x_i x_j$	$K(x_i, x_j) = (\gamma x_i x_j + r)^d$	$K(x_i, x_j) = \exp(-\gamma \lVert x_i - x_j \rVert^2)$
Hyperparameters	$C = 0.211$	$C = 1, \quad \gamma = 0.1$ $d = 3, \quad r = 0$	$C = 10^6$ $\gamma = 1.44$
Margin Type	Low value of C ∴ soft margin i.e., allows some misclassifications for better generalisation.	Moderate value of C ∴ soft margin, but stricter than SVM_linear.	High value of C ∴ hard margin i.e., the model is penalised heavily for any misclassifications.
No. of Support Vectors	3 500	6 000	15 000
Sensitivity	Train: 0.9354 Test: 0.9403	Train: 0.9418 Test: 0.94749	Train: 1.000 Test: 0.7351
AUC	Train: 0.9001 Train: 0.8979	Train: 0.9198 Test: 0.9174	Train: 1.000 Test: 0.8568
Comment	There is less than a 1% difference between the train and the test set for both metrics, therefore the model has **low variance**. Both metrics are above 90%, therefore the model has **low bias**. The SVM_linear **model performs well**.	There is less than a 1% difference between the train and the test set for both metrics, therefore the model has **low variance**. Both metrics are above 90%, therefore the model has **low bias**. This model has the highest AUC and sensitivity scores, making it the **best performing one**.	There is a 15% to 25% difference between the train and test set with both metrics, indicating **high variance**. Both metrics are above 70%, which is **fairly good** for **bias**. This model is **overfitting** and the **worst performing one**.

*Please note that due to excessive run times, **SVM_poly** and **SVM_radial** were performed on smaller subsets of the data*

Neural Networks

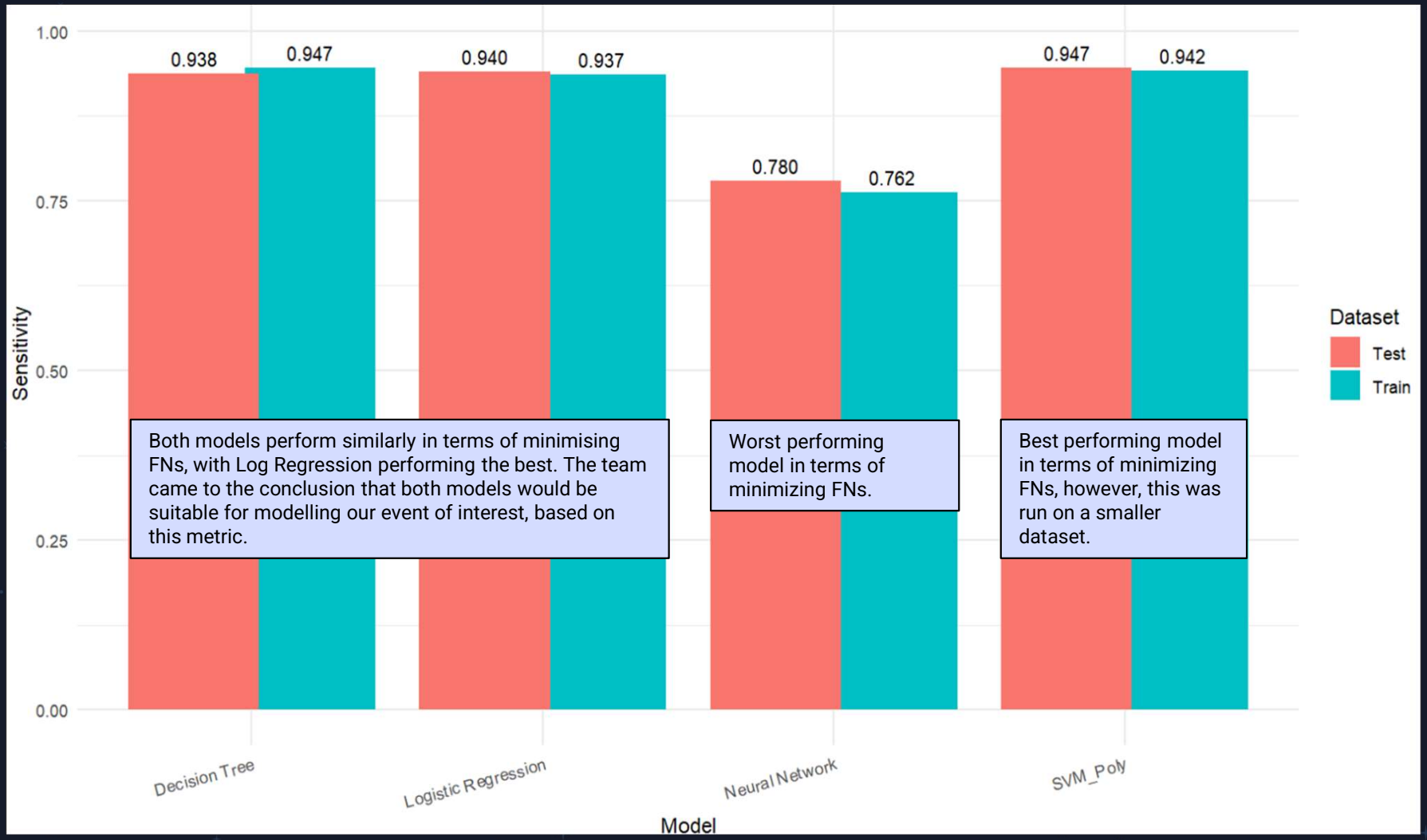


Neural Networks



- NNs is consistent with other ML techniques in that it detected absolute_magnitude_h as a critical variable in determining if an asteroid is potentially hazardous or not
- orbit_range_class_type remains a critical factor, as it defines if the object is orbiting inside Earth's orbit, outside earth's orbit or if they cross Earth's orbit (the most dangerous type of objects to Earth, are objects which cross Earth's orbit).
- observation_duration_days was picked up as important here, where in other ML techniques it was not picked up.
- avg_diameter is seen as the least important, after absolute_magnitude_h, orbit_class_type etc. has been taken into account

Comparison & Selecting Best Performing Model: Sensitivity

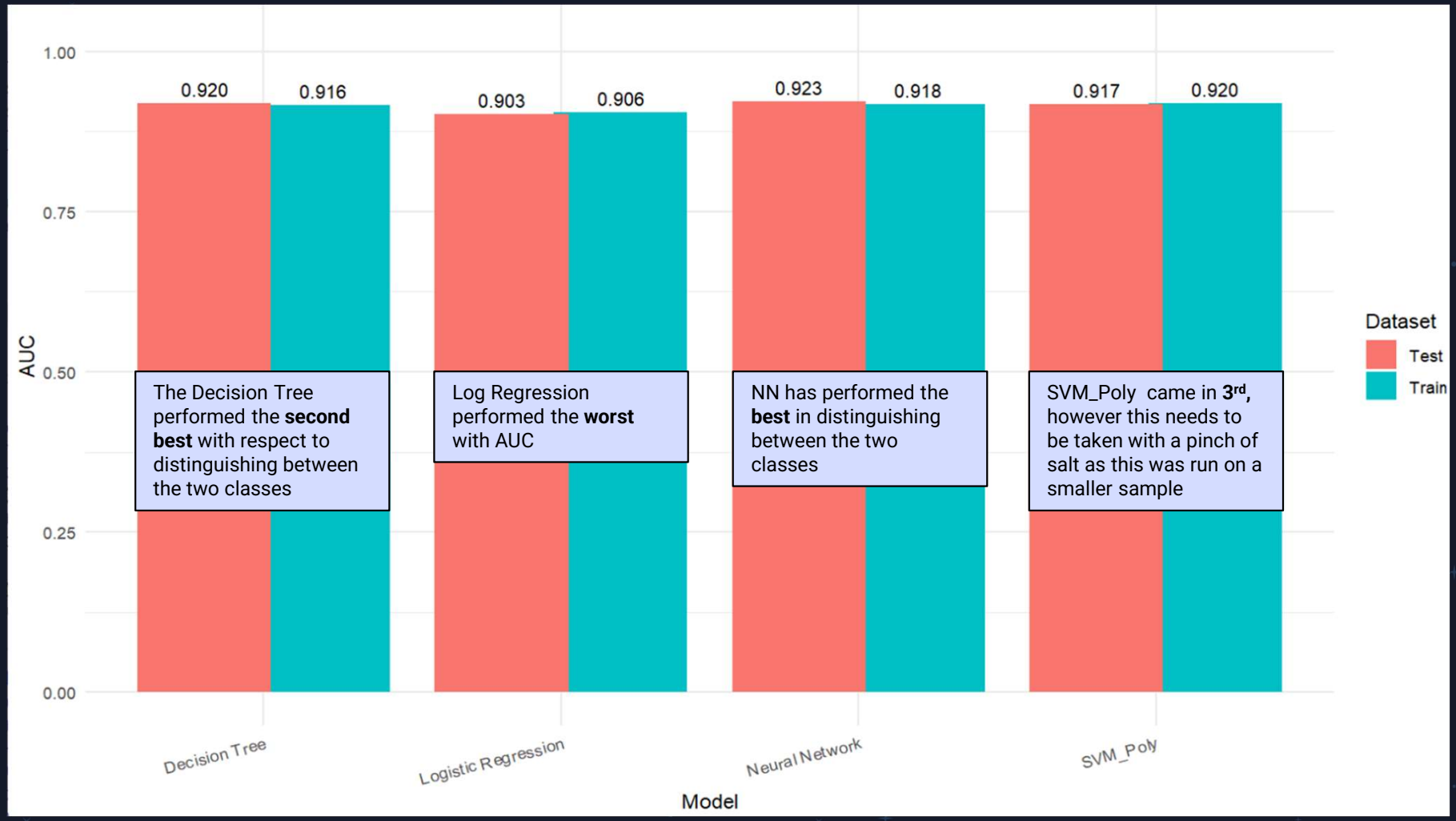


Small differences between train and test sets therefore the models all exhibit low variance

NN exhibits the most bias (slightly underfitting)

The other models exhibit low bias with sensitivity scores above 93%

Comparison & Selecting Best Performing Model : AUC



The Decision Tree performed the **second best** with respect to distinguishing between the two classes

Log Regression performed the **worst** with AUC

NN has performed the **best** in distinguishing between the two classes

SVM_Poly came in **3rd**, however this needs to be taken with a pinch of salt as this was run on a smaller sample

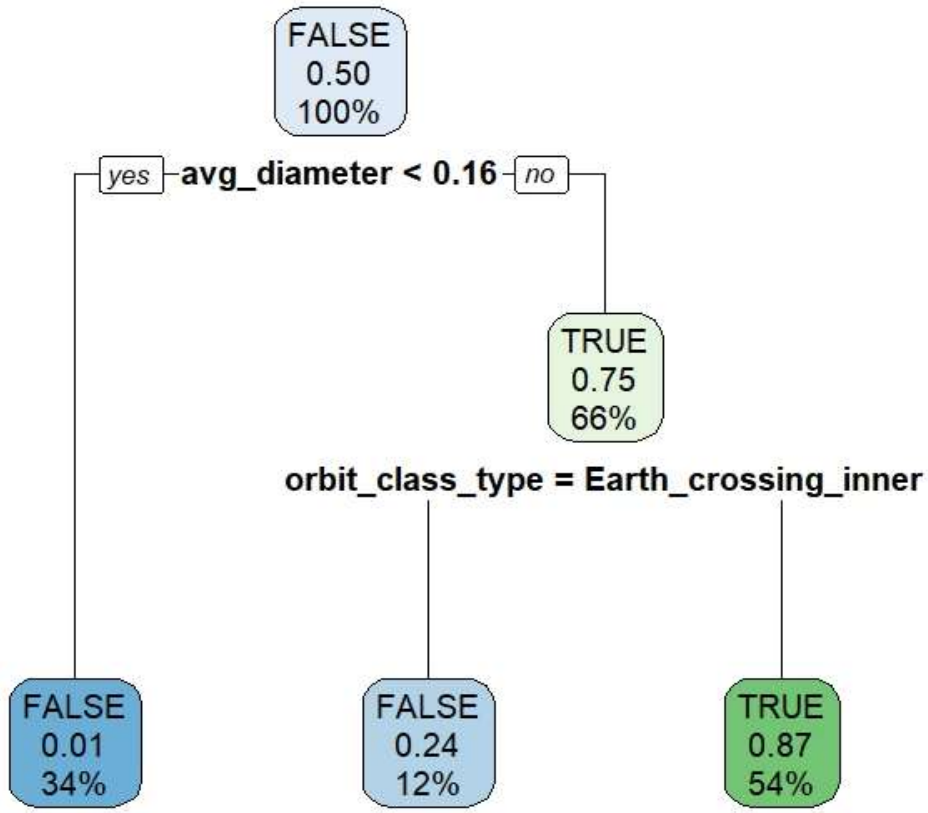
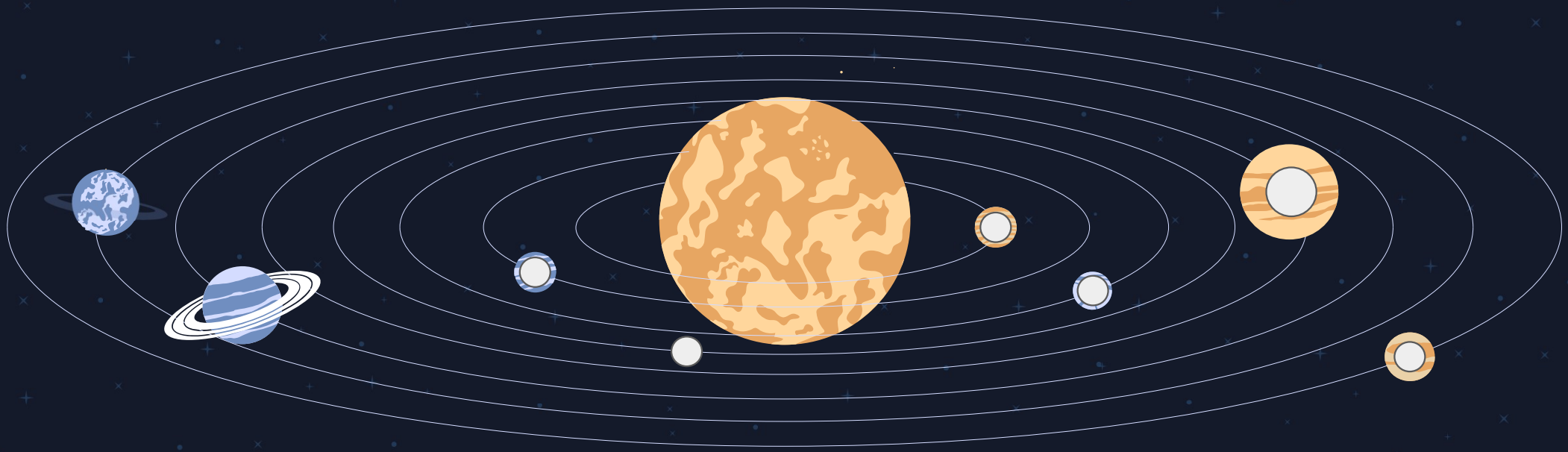
Small differences between train and test sets therefore the models all exhibit low variance

AUC > 90% therefore all models exhibit low bias for this metric and are fitting well

Selecting Best Performing Model

- NN had the best AUC, but the worst sensitivity score on both the training and the test set (below 80%), therefore the team did not feel this was an adequate model.
- SVM_poly performed well with both sensitivity and AUC, however since it was run on a **smaller sample**, the team did not feel confident with how it would perform on the full sample set. Therefore, this model was not selected as the best performing model.
- Logistic Regression performed well with a sensitivity score of 94% on our test set, however it had the lowest AUC score on the test set at 90.1%, therefore it was not selected as the best performing model.
- The Decision Tree had a sensitivity score of 93.8% and an AUC of 92%, therefore the team has selected the **decision tree as the best performing model**
- Overall, most models fit the data well, indicating that the data cleaning and feature engineering performed improved predictability

Winning Model



Thank You